

Lecture 1 · Preference, Choice and Demand

MWG Chapters 1 and 2 · two routes to the same theory, and what each one buys

Advanced Microeconomics 1

Monday 21 September

Before any economics: twenty minutes of decisions

The Session 1 quiz

- Open the course page and follow the **quiz link** [link / QR here]. Work alone, in silence; about 20 minutes.
 - Around thirty short questions: allocation problems and a few quick choices. **There are no right answers** — choose what you actually prefer.
 - You will invent a **random ID** on the first screen — *write it down*, you will need it again. Responses are anonymous, ungraded, and you may skip the whole thing without penalty. [Optional: one decision, for one randomly chosen participant, will be paid for real.]
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- Your answers become a dataset that this course will return to more than once this term. That is all I will say about it today.

This course in one arc



- Every block of theory in this course is the engine of a live empirical literature.
- Problem sets stay ruthlessly technical: the applications are why you tolerate the algebra, not a substitute for it.

Why start with axioms? Because rationality is measurable

- “Rational agent” sounds like an article of faith. It is not: consistency with utility maximization is a **testable, quantifiable property** of a finite dataset of choices.
- Humans: in a large representative Dutch panel, consistency scores are heterogeneous and correlate systematically with wealth, education, and age (Choi, Kariv, Müller, Silverman, AER 2014).
- Machines: instructed to allocate budgets, GPT’s choices were *more* consistent with utility maximization than human subjects’ (Chen, Liu, Shan, Zhong, Zhou, PNAS 2023).
- By Thursday week you will have computed these scores yourselves — the machinery starts now.

Chapter 1: two modeling routes

Primitives and two routes

Let X be the set of alternatives (bundles, careers, portfolios, ...).

Route A: preference-based

Start from a preference relation $\succsim$ on X ; impose *rationality axioms*; derive choice behavior.

Route B: choice-based

Start from observed choice behavior itself; impose *consistency axioms* (WARP); ask when it looks “as if” preferences exist.

- Route B treats the model's testable implications as primitives — the ancestor of modern revealed-preference econometrics.
- Central question of Ch. 1: **when are the two routes equivalent?**

Preference relations and rationality

Definition · preference relation

$\succsim$ is a binary relation on X ; $x \succsim y$ reads “ x is at least as good as y .” Derived:
 $x \succ y \iff x \succsim y$ and not $y \succsim x$; $x \sim y \iff x \succsim y$ and $y \succsim x$.

Definition · rationality (MWG 1.B.1)

$\succsim$ is *rational* if it is **complete** ($x \succsim y$ or $y \succsim x$, always) and **transitive** ($x \succsim y \succsim z \Rightarrow x \succsim z$).

- What they rule out: incomparability (completeness — strong for unfamiliar or morally loaded choices); cycles and framing effects (transitivity). An agent with cyclic strict preferences can be **money-pumped**: pay ε at each step of the cycle, forever.
- Consequences (Prop. 1.B.1): $\succ$ irreflexive and transitive; $\sim$ an equivalence relation. Proofs: PS1.

Utility representation

Definition · utility representation

$u : X \rightarrow \mathbb{R}$ represents $\succsim$ if $x \succsim y \iff u(x) \geq u(y)$.

- If some u represents $\succsim$, then $\succsim$ is rational (completeness and transitivity of $\geq$ transfer back).
- **Ordinality:** any strictly increasing $f \circ u$ represents $\succsim$ too. Levels carry no meaning; only the ranking does. Anything not invariant to monotone transformations is not (yet) economics — cardinal content returns only under uncertainty (Ch. 6).
- **Existence:** finite X — rationality suffices ($u(x) = \#\{y : x \succsim y\}$, PS1). Infinite X — rationality is *not* enough: lexicographic preferences on $\mathbb{R}^2$ are rational, strongly monotone, and have **no representation**. The missing ingredient is continuity; Debreu delivers existence next lecture.

Choice structures and WARP

Definition · choice structure

$(\mathcal{B}, C(\cdot))$: a family $\mathcal{B}$ of budget sets $B \subseteq X$ and nonempty chosen sets $C(B) \subseteq B$. Read: **the experiments we ran, and their outcomes**. No mental states assumed — which is why this travels to lab data, scanner data, and LLM outputs.

Definition · WARP (MWG 1.C.1)

If $x, y \in B$ with $x \in C(B)$, then for any B' with $x, y \in B'$ and $y \in C(B')$, also $x \in C(B')$.

- In words: if x is ever chosen with y available, y may never be chosen *over* x .
- Violation: $C(\{x, y\}) = \{x\}$ but $C(\{x, y, z\}) = \{y\}$. Not exotic — exactly what decoys, menu effects, and limited attention produce. WARP is where standard and behavioral economics part ways.

When the two routes coincide

Proposition · MWG 1.D.1

If $\succsim$ is rational, the induced choice rule $C^*(B, \succsim) = \{x \in B : x \succsim y \ \forall y \in B\}$ satisfies WARP.

Proof. $x \in C^*(B)$, $y \in B$, and $y \in C^*(B')$, $x \in B'$. Then $x \succsim y$ and $y \succsim z$ for all $z \in B'$; transitivity chains to $x \succsim z$ for all $z \in B'$, so $x \in C^*(B')$. $\square$

Proposition · MWG 1.D.2

If $(\mathcal{B}, C)$ satisfies WARP **and** $\mathcal{B}$ contains all subsets of up to three elements, then the revealed relation $\succsim^*$ ($x \succsim^* y$ iff x chosen with y available) is the unique rational preference rationalizing C .

- Why *three*? Transitivity is a statement about triples; with pairs only, WARP can hold while no rational preference exists (PS1 makes you build the example).

The gap that matters in practice

The empirical problem

Real consumption data give us only *budget sets* — a very poor $\mathcal{B}$. Pairs and triples are never observed, so WARP alone will not secure rationalizability in the market setting.

- Fix: strengthen WARP to rule out revealed-preference *chains* cycling back — **GARP**: no sequence $x^1 R x^2 R \dots R x^n$ with x^n strictly revealed preferred to x^1 .
- **Afriat's Theorem** (1967; stated and proved in Lecture 4): a finite dataset $\{(p^t, x^t)\}$ satisfies GARP *iff* it is rationalizable by a continuous, increasing, concave utility. For finite data, **utility maximization = GARP** — nothing more, nothing less.
- Thursday week's practical: you implement this equivalence, on real choices, before you ever see it proved.

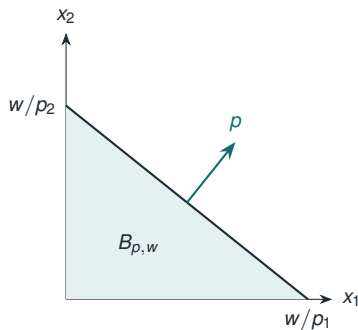
Chapter 2: the Walrasian environment

Commodities, consumption set, budget set

- L commodities, distinguished by traits, *location, date, and state of the world* — one definition that carries us quietly to GE and finance. Consumption set $X = \mathbb{R}_+^L$ (convex; rules out indivisibilities and survival constraints).
- **Walrasian budget set**, $p \gg 0$, $w > 0$:

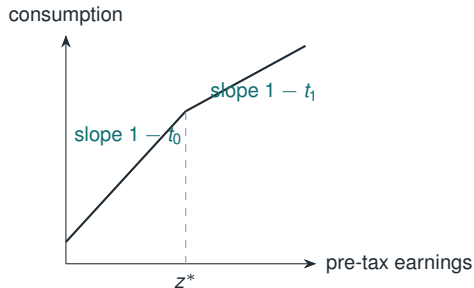
$$B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}.$$

- Price taking; linearity (every unit at the same price); boundary a hyperplane with normal p ; convex — inherited by everything downstream.

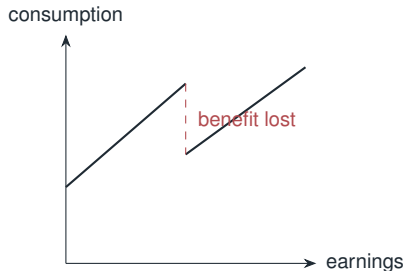


Real budget sets are not hyperplanes

Kink (e.g. progressive tax)



Notch (benefit cliff)



- Progressive taxes, EITC phase-outs, benefit cliffs: piecewise-linear, sometimes non-convex budget frontiers.
- Hold the thought: choice theory predicts *exactly what happens at these corners*, and that prediction becomes an identification strategy for elasticities in Lecture 4.

Walrasian demand and two accounting laws

Definition · Walrasian demand

$x(p, w) \subseteq B_{p,w}$: the chosen bundle(s) at each price–wealth pair; assume single-valued and continuous for now.

Two assumptions maintained throughout Ch. 2 (both *derived* from preferences in Ch. 3):

1. **Homogeneity of degree zero**: $x(\alpha p, \alpha w) = x(p, w)$ — only relative prices and real wealth matter; “no money illusion.” Practical use: normalize.
2. **Walras’ law**: $p \cdot x(p, w) = w$ — full expenditure.
 - Differentiating the two identities yields free cross-equation restrictions on any demand system — **Engel aggregation** ($\sum_{\ell} p_{\ell} \partial x_{\ell} / \partial w = 1$: a marginal dollar is spent somewhere, so not every good can be inferior) and **Cournot aggregation**. You derive both on PS1 and verify them on CES.

WARP restated for market data

Definition · WARP in the market (MWG 2.F.1)

$$p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w) \implies p' \cdot x(p, w) > w'.$$

- Read: if the bundle chosen at (p', w') was *affordable* at (p, w) but not chosen there, then when it *was* chosen, the other bundle must have been unaffordable. Otherwise the two observations contradict each other.
- One axiom about pairs of observations. Its power is unlocked by asking a well-chosen pair — next slide.

Slutsky compensation and the law of demand

Definition · compensated price change

From (p, w) with $x = x(p, w)$, move to p' and set $w' = p' \cdot x$: wealth adjusted so *the old bundle remains exactly affordable* — the budget line pivots through the original choice, stripping out the mechanical income effect *using only observables*.

Proposition · MWG 2.F.1

Under h.d.0 and Walras' law: $\text{WARP} \iff$ for every compensated change,

$$(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0, \quad \text{strictly if } x(p', w') \neq x(p, w).$$

Proof ($\Rightarrow$), the four lines to know cold. With $x' \neq x$:

$$(p' - p) \cdot (x' - x) = \underbrace{p'x'}_{=w'} - \underbrace{p'x}_{=w'} - p x' + \underbrace{px}_{=w} = w - p x'.$$

x is affordable at (p', w') yet x' was chosen: WARP forces $p x' > w$. $\square$

The Slutsky matrix, built from choices

Take dp small, compensate $dw = x \cdot dp$; the law of demand becomes $dp \cdot dx \leq 0$ with $dx = [D_p x + D_w x x^T] dp$.

Definition · Slutsky substitution matrix

$$S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^T, \quad s_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k.$$

- Every entry is estimable from demand data alone — **no utility function anywhere**. This object, built here from consistency axioms, will be rebuilt from preferences in Lecture 3, and the comparison of the two constructions is the deep structure of the course's first half.

The scorecard: what WARP buys, and what it doesn't

Proposition · consequences (MWG 2.F.2–2.F.3)

If x is h.d.0, satisfies Walras' law and WARP, then $S(p, w)$ is

- **negative semidefinite**: $dp \cdot S dp \leq 0$;
- **singular**: $S p = 0$ and $p^T S = 0$ (from homogeneity + Walras).

What WARP does not buy

For $L \geq 3$, WARP does **not** imply S **symmetric** (MWG Example 2.F.1). Symmetry is the extra content of *transitivity* — of one coherent preference behind all choices — delivered by full preference maximization in Lecture 3, and the key to recovering preferences from demand in Lecture 4.

- Keep the ledger: NSD + singularity = pairwise consistency. Symmetry = no cycles anywhere.

Takeaways and readings

Takeaways

- Rational $\succsim \Rightarrow$ WARP always; the converse needs a rich domain (all sets up to size 3) — and market data are poor, so finite data will need GARP and Afriat.
- Two accounting laws (h.d.0, Walras) already restrict demand derivatives; WARP $\iff$ the compensated law of demand.
- $S = D_p x + D_w x x^T$ is NSD and singular from consistency alone; *symmetry is not implied* — that is transitivity's fingerprint, coming Thursday.

Required: MWG Ch. 1 and Ch. 2 (all); Example 2.F.1 carefully.

Applications: Chen et al. (PNAS 2023); Choi et al. (AER 2014), intro + Table 1.

Ahead: the practical is a week Thursday — all you need is a laptop with a working R installation. Check it starts *this week*, not that morning.